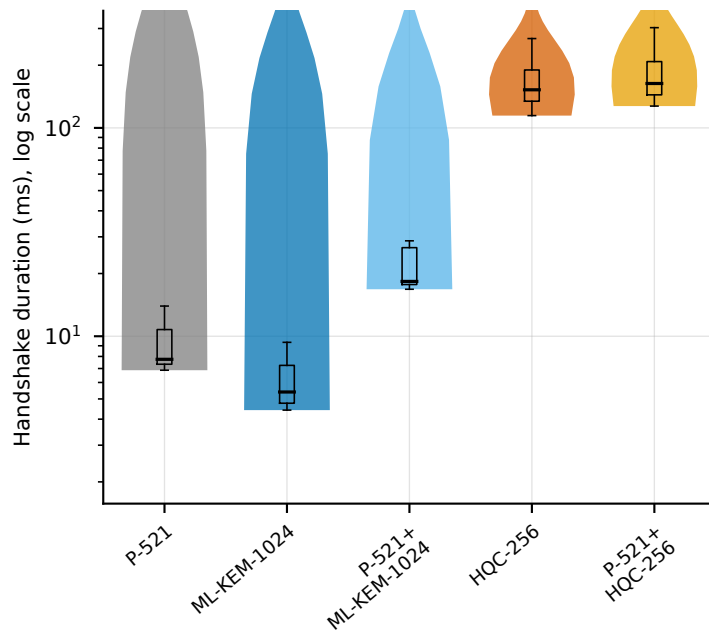
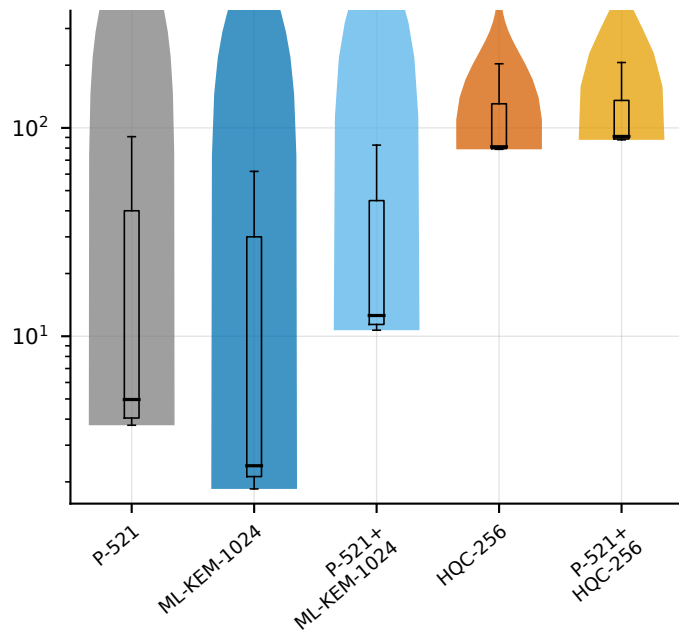


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)	mldsa87 Fail
P-521	413.3	258.0	7.00k	4/500	733.8	218.1	7.09k	13/500
ML-KEM-1024	277.8	279.7	7.00k	0	497.6	246.8	7.11k	1/500
P-521+ML-KEM-1024	249.6	237.2	7.01k	0	574.8	232.3	10.19k	2/500
HQC-256	245.8	124.6	3.09k	0	211.1	165.2	3.10k	0
P-521+HQC-256	286.0	146.5	3.19k	1/500	249.4	212.0	7.06k	0